

CMPPM125 - Assignment 5 Report

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Expanded Base Game:

Our group decided to expand the **Curling Physics Game from Assignment 2** to turn it into a unique mini golf experience. It includes 10 unique courses, each with obstacles to avoid and a scoring system based on curling. The game supports 2 players locally, each taking turns to move their curling stone using movement based on the Curling Game assignment to land nearest to the center of the scoring zone in the least amount of launches.

Controls:

Hold left click to build up power

Right Click + drag = change camera perspective

Left Click + drag = change aim direction

Left Click + drag on directional arrows = curving stone (only after thrown)

Summary of Changes:

We began with taking the original curling game created by Michael Tang and Vivian Kim in assignment 2. With the goal of rewriting the game to work as a mini golf game, we completely rewrote the player turn systems and scoring systems. The scoring system now has each player control only a single curling stone rather than several. This acts as the player's "golf ball." Rather than each player only getting one launch per stone, they are able to launch as many

times as they please **until they touch the target zone. If they leave the target zone after entering, either by overshooting** the zone or being knocked out by an opponent's shot, they forfeit any additional turns.

The scoring mixes both curling's base scoring with mini golf's scoring style. Thus, players earn points based on several factors:

- Players earn points based on their **position inside the target zone**. They earn more points near the center and none if they are not within the zone at all.
- Players earn bonus points for being the **nearest player to the center**. This occurs even if neither player is actually within the scoring zone by the end of the course.
- Players earn bonus points for reaching the scoring zone in the **least amount of launches**.

The score is added up and the player with the highest score by the end of the 10 courses is crowned the winner.

Next, in order to create a more exciting mini golf experience, we needed to introduce various obstacles. We added 4 unique obstacles that are sprinkled throughout the courses:

- A **bumper obstacle** that pushes the player in the opposite direction if they touch it. These come in different sizes and with differing strengths.
- A **moving obstacle** that takes several objects and evenly distributes them to move along a spline path at a given speed.
- A **rotating obstacle** that is able to turn at any speed in any direction.
- A **thrower enemy obstacle** that follows a determined spline path. It features a vision radius, and if the player enters their sight, the thrower will chase after the player. If the player's stone is caught by the thrower, they will be picked up and thrown toward a point, setting them back significantly. Otherwise, if the thrower moves too far from their path, they will give up on the player and return to patrolling.
 - Thrower enemies each use nav meshes to ensure that they accurately interact with obstacles in their path.

Next, we changed the original single curling course into **10 unique mini golf-like courses**. Each course has its own spawn points and target zones. The first course is simple and mimics the original curling field to allow players to get used to their controls, but as the player nears the end of the game, courses become more complex and more difficult.

Once all players have completed all courses, the game ends and players are sent to the game results screen, showing the winner and each player's scores.

Additionally, we added a basic title screen that players are greeted with upon entering the game.

We also added sound effects that can be heard on button clicks, when curling stones are sliding, and when sweeping to curve the stone's trajectory. There is also music in the background to add to the atmosphere.

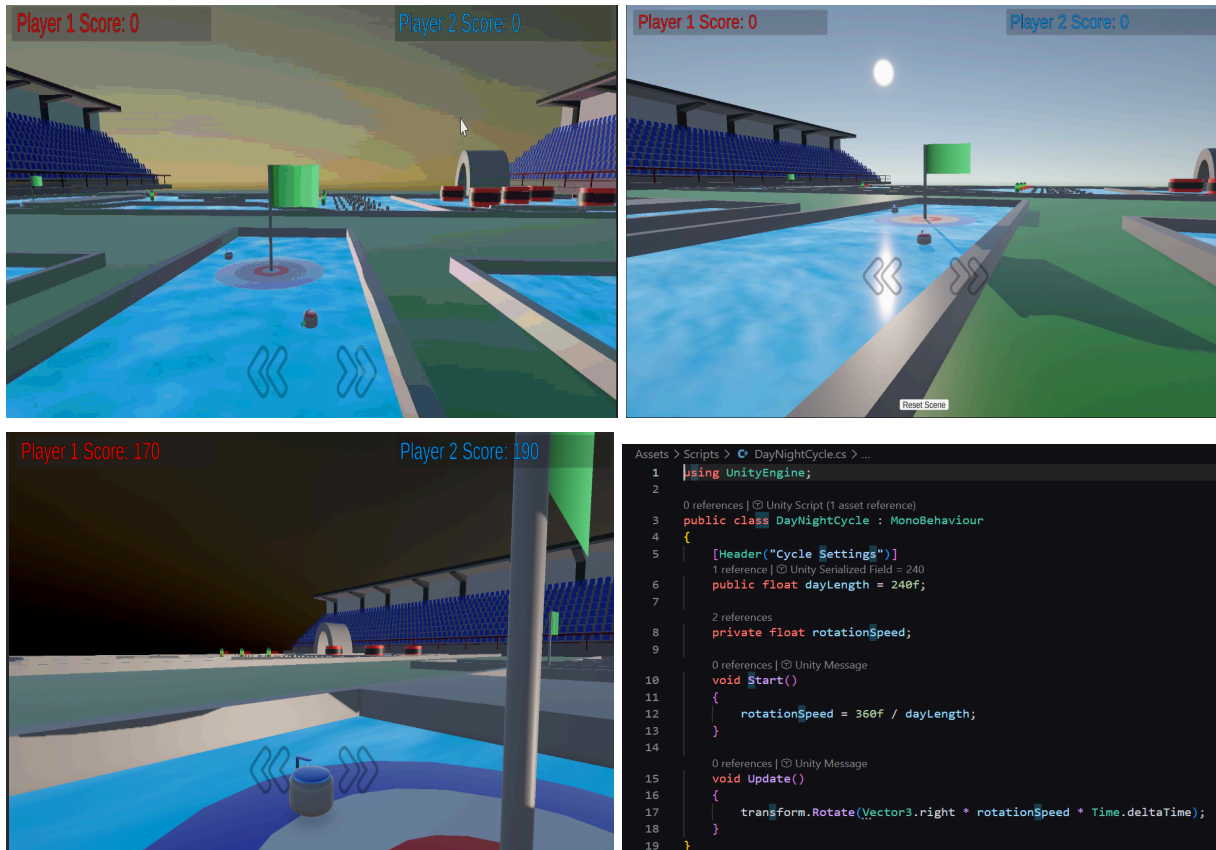
Additionally, we added various visual additions that are detailed in the "Graphics Integration" section below. There are also custom camera controls to allow the player's camera to orbit around their curling stone. They are able to zoom in and out, and it dynamically moves toward

the target between turns and courses. Simply hold right-click and drag to turn or use the mouse wheel to zoom.

Graphics Integration:

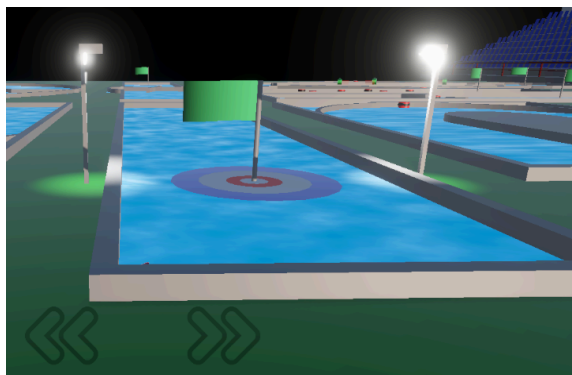
Day/Night Cycle

We added a day/night cycle to our game to create some dynamic lighting. Using a custom MonoBehaviour Script, we created a simple day-night cycle effect by rotating the directional light



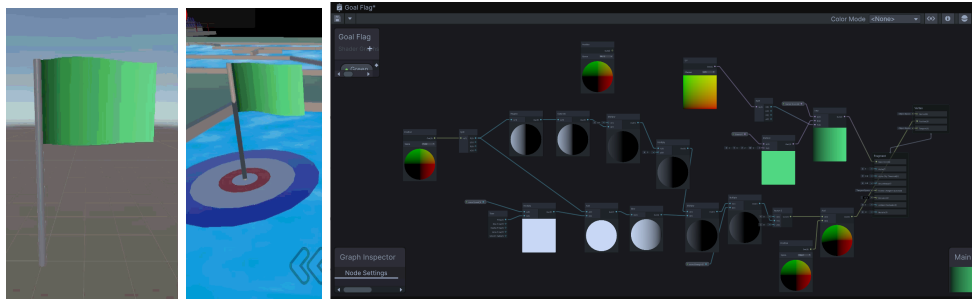
Street Light Point Lamps:

We added street lamps in the first level as a way of lighting up the arena when it gets dark.



Flowing Flag Shader Graph:

In addition to adding a day-night cycle, we also added a custom scoring flag shader graph. This shader graph incorporates a sine wave function that is mapped to a plane's UV coordinate positions in order to simulate a flowing flag. We also added a green gradient color to the flag to mimic golfing flags.



Audio Assets:

List of all audio assets used with attributions:

Audio Assets were created by **Leonidas Assimes** using beepbox.co.

This includes:

- UI clicking sound effect
- Curling stone moving sound effect
- Curling stone sweeping (turning) sound effect

The background music is a royalty-free track created by *Alba_Mac* on [pixabay](https://pixabay.com).

Key Features:

Screenshots:

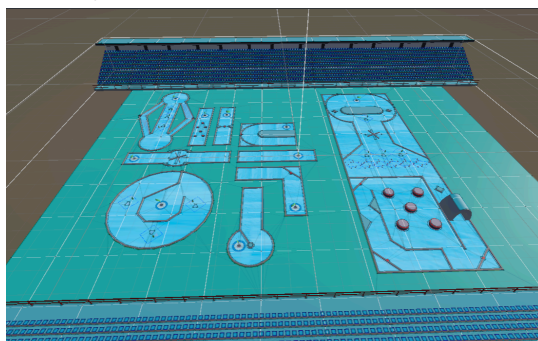
End screen:



End of Round score:



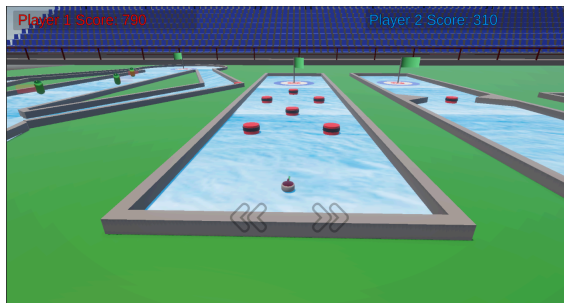
10 Curly Golf courses:



2-player action:



Player at start of course:



Short Reflections:

Michael Tang:

My role in this project was to take the already existing curling game and refactor its core game mechanics to work in a 2-player golf game. Since the base curling game was from my Assignment #2, I undertook the process of refactoring the underlying code systems to work for a mini golf-style game. This process involved completely redesigning the game's turn-based system to allow the player to continue where their previous curling stone landed. Instead of respawning the current player's stone back at the spawn location, I had to rewrite the code logic such that after every completed hole, the spawn location and current active scoring zone changed to the next hole. Additionally, for this assignment, I was responsible for redesigning the scoring target zone to provide points based on how close the player was to the center of the zone. Bonus points were also added for being the closest player to the center and being the player who used the least amount of shots to get to the zone. I was also responsible for creating the game's Shader Graph Flag that would mark the scoring target.

Although the base curling game was already created, refactoring the code to work within the bounds of our project posed a significant challenge. There were many times where entire sections of the game's turn manager script had to be completely rewritten to accommodate the new game logic. Likewise, I had difficulties in using the trigger collider on the scoring target as there was an issue at one point where the game would record the player's position incorrectly within the scoring zone.

Over the course of this project, I learned so much about working within the Unity Engine. I personally had a great time learning how to incorporate collision detection, trigger colliders, and Custom Shader Graphs together to create a working Unity game project. I think that if I had more time to work on this, I would have liked to have added more shader graph assets. Aspects like a flowing water fall, burning pits or lava, and even a spinning windmill would have been an interesting challenge that would have added a significant amount of polish to the different levels. I would have additionally enjoyed adding in a how-to-play screen that would give the players a step-by-step tutorial on how to play the game.

Andrew Degan:

2-3 paragraphs on your contributions to the project, technical challenges you overcame, what you learned throughout the quarter, and what you would do differently with more time.

My primary role in this project was to create the stage obstacles. I created the bumper obstacle, which detects a collision from the player's curling stone and then applies a force to launch them in the opposite direction. The force multiplier allows for different bumpers to apply different forces to create the effect of different bumper strengths. I also added the moving obstacle, which takes a spline object as a path and a prefab of your choice. It spawns a defined number of the obstacle of your choice, evenly spaced, to move around the path all at once. Another obstacle I created was a rotating obstacle, which very simply rotates an object at any speed in any direction. Lastly, I created an obstacle that utilizes AI, being the thrower enemy obstacle. This obstacle patrols a set spline path and features a visual vision radius trigger. If the player enters the obstacle's vision radius, then the thrower will chase after the enemy, utilizing a navmesh to ensure that it correctly traverses the landscape. If it catches the player, they will be picked up and thrown toward a point. If the obstacle runs too far from their path, they will give up and return to patrol.

I additionally took on the role of implementing the sound effects into the game and finding/adding music. Each button or score menu click plays a UI click sound, the curling stones sliding have dynamic audio that gets louder depending on the stone's speed, and a sweeping sound plays when the curling stone turns. All sound effect audio was created by Leo. I also made various changes to the project for quality assurance and created the orbital camera system to allow for camera zooming, rotating, and smooth movement.

My main technical challenges were creating the thrower obstacle, as it utilized AI against a target that is usually moving quickly. I also had some trouble implementing the camera, as sometimes it would not center on the target correctly or could become jittery as it moved. Over this course, I learned many useful skills using Unity, especially with lighting and environment design. Though I did not showcase them on this particular project, I found that it was great for creating intricate environments that add a lot of player intrigue. I also gained much more knowledge and experience creating game AIs, especially with state machines, which will come in handy in several projects. If I had a bit more time to work on this project, I would likely work on fledging out the environment more. I think that the gameplay itself is good, but the lighting and atmosphere could be more interesting if we had more assets to work with.

Aiden Le:

I created the 10 courses for our game, using the bumper, thrower, and scoring field prefabs and adding them to the game manager. I also created a quick day/night cycle for our game with each day being 4 minutes long. I also created a quick stadium to house the courses making it look like a very simplified olympic style area.

The technical challenges I faced was figuring out how to use probuilder in unity. It can be really helpful to create complicated shapes, but learning how to was the hardest part. Also learning how to create the thrower's path was a little bit of a problem. But overall it was quite easy to learn everything.

Throughout this quarter I think the most important thing I learned was AI states and just getting used to working in 3d unity. I've only worked with 2d unity before and learning 3d by myself was a little daunting, so this class really helped me get comfortable with it. If I were to take this class again, I would like to take it without having other project heavy classes so I could focus on this

one. I would like to do this class with my full focus and see what I could create.

Leonidas Assimes:

I created the main menu and linked up the buttons to the appropriate scenes. This includes a start button, an options button to control sound volume, a credits viewer, and a quit button. I also created sound effects and found music for our game. I used beepbox.co to create sound effects with instruments in order to create the most accurate portrayal of a curling stone sliding on ice. Additionally, I added lamp post objects to the scene to allow for more dynamic lighting, especially at night. Most of the challenges for me were not technical, but from mental blocks and burnout over the years. I did encounter some issues with scaling the menu for different screen sizes, but proper testing sorted everything out. Over the quarter I learned a ton about using Unity, both from a technical and mental perspective. It turns out Unity isn't as daunting as it seems when you just jump into it. I wish I had more time in the quarter to just mess around with Unity.